

NIGERIA ENERGY TRANSITION BRIEF

2025 EDITION

What rising fuel costs, the CNG opportunity, solar economics, biogas potential, and DFI financing mean for Nigerian operators planning their next energy capital decision.

■ 1,200+

Petrol price/litre
2025 average

~60%

Potential cost saving
with CNG conversion

\$2.1B

CNG infrastructure
gap 2024–2030

8M mt

Annual agri-waste
CBG feedstock potential

A CircularEnergies Advisory Publication · circularenergies.africa · advisory@circularenergies.africa

ABOUT THIS BRIEF

The Nigeria Energy Transition Brief is a free annual publication from CircularEnergies — an energy transition feasibility advisory firm headquartered in Abuja, Nigeria. We serve Nigerian SMEs and energy operators across logistics, agribusiness, manufacturing, hospitality, and energy distribution who are evaluating cleaner, more cost-effective fuel and energy alternatives.

This brief is designed to give decision-makers a concise, data-informed overview of where Nigeria's energy transition stands in 2025 — and what it means for operators planning their next capital move. All data reflects publicly available Nigerian sources including NNPC, NERC, CBN, the Nigeria Midstream and Downstream Petroleum Regulatory Authority (NMDPRA), and international energy agencies where relevant.

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Disclaimer: This brief is for informational purposes only and does not constitute investment, financial, or professional advisory advice. Data is sourced from publicly available information and CircularEnergies' internal research. Operators should commission a project-specific feasibility study before making capital investment decisions.

The Nigerian Energy Cost Crisis — Where We Stand in 2025

Nigeria's energy cost landscape has been fundamentally restructured since the removal of petrol subsidies in May 2023. What was once an artificially suppressed operating cost has become the single largest overhead for most Nigerian SMEs — and it continues to rise.

■1,200+	280%	■800–■1,400
Average petrol price per litre (2025)	Pump price increase since subsidy removal (May 2023)	Diesel price range per litre across Nigeria (2025)

The impact is not uniform across sectors. Logistics and transport operators — who depend directly on petrol and diesel for fleet operations — have experienced the most acute margin compression. Manufacturing and agribusiness facilities relying on diesel generators for power backup have seen energy costs rise from 15–20% of operating expenditure to 35–45% in many cases.

ENERGY COST AS % OF OPEX — BEFORE AND AFTER SUBSIDY REMOVAL

Sector	Pre-2023 Energy % OPEX	2025 Energy % OPEX	Key Driver
Logistics & Transport	18–22%	38–48%	Fleet petrol/diesel
Manufacturing	15–20%	32–42%	Generator diesel
Agribusiness / Food Proc.	12–16%	28–36%	Processing + cold chain
Hospitality (Hotels)	20–25%	40–50%	Generator + cooling
Commercial Buildings	10–15%	25–35%	Generator fuel

Source: CircularEnergies analysis, NNPC pricing data, operator cost surveys 2024–2025.

"The question for Nigerian operators in 2025 is no longer whether to consider alternative energy — it is which alternative makes financial sense for their specific operation, and how to finance the transition."

CNG: The Most Accessible Transition for Fleet Operators

Compressed Natural Gas (CNG) represents the most commercially mature and immediately accessible alternative fuel pathway for Nigerian fleet operators. Nigeria's vast gas reserves, the federal government's Presidential CNG Initiative (PCNGi), and the rapidly expanding retail CNG station network make this the most credible near-term transition for logistics, transport, and commercial vehicle operators.

CNG vs Diesel/Petrol — Cost Comparison (2025)

Parameter	Petrol / Diesel	CNG	Saving
Fuel price equivalent (per litre equivalent)	■1,100 – ■1,400	■250 – ■400	60–75%
Typical monthly fuel cost (40-truck fleet)	■8M – ■12M	■2.5M – ■4M	■5M – ■8M/mo
Engine conversion cost (per vehicle)	N/A	■350k – ■600k	Payback: 4–8 months
Maintenance cost differential	Baseline	~15% lower	Cleaner combustion
CNG station availability (Nigeria)	—	200+ retail pts	Growing rapidly

Note: CNG price data reflects PCNGi-regulated stations. Private station pricing may vary.

Who Should Be Looking at CNG Right Now?

- Logistics and haulage companies operating 10 or more vehicles on regular routes
- Bus and mass transit operators in Lagos, Abuja, Kano, Port Harcourt, and Ibadan
- Manufacturers running on-site vehicle fleets (forklifts, shuttles, delivery trucks)
- Government agencies and parastatals with fleet management mandates
- Agricultural commodity aggregators running collection vehicle fleets

What a CNG Feasibility Study Gives You

A CircularEnergies CNG Feasibility Study analyses your specific fleet — vehicle count, daily mileage, route patterns, and fuelling behaviour — and produces: (1) a vehicle-by-vehicle conversion cost model, (2) monthly fuel savings projection under current and stressed CNG price scenarios, (3) IRR and payback period for the full conversion investment, (4) nearest CNG infrastructure assessment, and (5) a Go/No-Go recommendation with financing options. Starting from ■400,000. Delivered in 2–3 weeks.

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Solar + Storage: Making the Numbers Work for Industry

Solar hybrid systems have crossed the commercial viability threshold for Nigerian industrial and commercial operators. Falling panel prices, improving battery economics, and Nigeria's exceptional solar irradiance (4.5–6.5 kWh/m²/day across most of the country) now mean that for facilities spending ₦500,000 or more per month on diesel generation, a solar hybrid system typically pays back within 3–5 years — with a 20-year asset life.

Parameter	Diesel Generator	Solar Hybrid System
Cost per kWh generated	₦180 – ₦280	₦35 – ₦65 (after CAPEX)
Typical CAPEX (100kW system)	N/A	\$80,000 – \$130,000
Monthly fuel saving (100kW equiv.)	Baseline	₦600k – ₦1.2M/month
Payback period	N/A	3–5 years typical
System lifespan	8–12 years (engine)	20–25 years (panels)
CO2 reduction	Baseline	70–90% vs diesel

Key financing options: CBN Renewable Energy Finance Facility, BOI Green Finance, IFC MSME Energy Program, and commercial bank green loans. Solar Feasibility Studies from ₦380,000.

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Compressed Biogas (CBG): Nigeria's Hidden Feedstock Opportunity

Nigeria generates an estimated 8 million metric tonnes of agricultural waste annually — from rice husks, cassava peels, maize stalks, livestock manure, and food processing residue — much of it currently burned or left to decompose. For operators sitting on this waste stream, anaerobic digestion and CBG production represent an opportunity to convert a disposal cost into an energy revenue stream.

State/Region	Primary Feedstock	Est. Annual Volume	CBG Potential
Kano / Jigawa	Groundnut, grain waste	800,000 mt	High
Benue / Nasarawa	Cassava, yam peels	650,000 mt	High
Cross River / Akwa Ibom	Palm, plantain waste	500,000 mt	Medium-High

Ogun / Oyo	Cassava, poultry litter	600,000 mt	High
Kaduna / Niger	Livestock manure, grain	700,000 mt	High

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LPG: The Clean Cooking Transition and Supply Chain Opportunity

Nigeria is Africa's largest LPG producer yet one of its lowest per-capita consumers. Over 80 million Nigerians still cook with firewood or kerosene. The federal government's National LPG Expansion Plan targets 5kg cylinder penetration across all states — creating a supply chain investment opportunity at every stage from refinery extraction to last-mile bottling and distribution.

LPG Supply Chain Investment Entry Points

Stage	Entry Point	Capital Required	Est. Margin
Extraction & fractionation	JV with refinery offtake	High (\$M+)	N/A (upstream)
Bulk storage & import	Coastal storage terminals	\$2M – \$10M	■15–■25/kg
Wholesale distribution	Bridge tank trucks	■80M – ■200M	■20–■40/kg
Bottling plant (LPG depot)	Franchise / independent	■15M – ■60M	■30–■60/kg
Retail last-mile	Cylinder retailers / agents	■2M – ■8M	■40–■80/kg

LPG Supply Chain Feasibility Studies from ■420,000. Covers bottling plant economics, catchment area demand modelling, NMDPRA licensing pathway, and 5-year financial projections.

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DFI Financing: What Nigerian SMEs Can Actually Access in 2025

One of the biggest barriers to energy transition investment in Nigeria is not willingness — it is knowledge of what financing is available and how to access it. Several DFI and government financing windows are currently open to Nigerian SMEs undertaking clean energy transitions, but most operators are unaware of them or unable to meet documentation requirements.

Facility	Provider	Ticket Size	Rate	Best For
CBN REAP (Renewable Energy)	CBN via PFIs	■10M–■500M	5–9% p.a.	Solar, biogas
BOI Green Finance	Bank of Industry	■5M–■200M	5% p.a.	Solar, CNG, LPG
IFC MSME Energy	IFC via local banks	\$50k–\$2M	Mkt rate	Solar, efficiency

NASENI CNG Fund	NASENI/NNPCL	Varies	Concessional	CNG conversion
AfDB / IsDB Lines	Via Nigerian DFIs	\$100k+	Concessional	Larger projects

Note: All financing facilities require a bankable feasibility study as part of the application. CircularEnergies' Standard and Comprehensive study packages include DFI financing documentation.

How to Know if a Feasibility Study is Right for You

A feasibility study is not always the right next step. It makes sense when you are seriously considering a capital investment in an energy transition, need to justify the decision to a board, bank, or business partner, or want to know the numbers before committing to a vendor or technology. Here is a quick guide:

You SHOULD commission a study if...	You may not need one yet if...
• You spend ₦500,000+ per month on diesel or petrol • You are considering converting 5+ vehicles to CNG • You need a bank or DFI to co-finance the investment • You have agri-waste you want to monetise • You want to enter the LPG distribution chain • A board or investor is asking for financial justification	• You are just beginning to explore options • Your monthly energy spend is below ₦200,000 • You have no timeline for a capital decision • You are not yet in a position to implement findings In these cases, start with a free consultation — we will tell you honestly

Ready to find out if a transition makes financial sense for your operation?

Contact CircularEnergies for a free 30-minute consultation. We will review your situation, tell you which study applies, and send you a fixed-fee proposal within 48 hours.

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Our Studies: CNG Fleet & Infrastructure · Solar + Storage · Biogas / CBG · LPG Supply Chain · Sustainable Aviation Fuel · Green Hydrogen Positioning